Solution:

Given $C(x)=\frac{x^2}{6}+5x+200$ … (1)

and $p(x) = 40-x$ …(2)

Profit is maximized when,
marginal revenue = marginal cost.

(i.e) $R'(x) = C'(x)$

$= \frac{x}{3}+5$

$R = p \cdot x$

$= 40x- x^2$

$R'(x) = 40-2x$

Hence $40-2x=\frac{x}{3}+5$

$x = 15$

At $x = 15$, the profit is maximum.

6.2.3 Local and Global (Absolute) Maxima and Minima

Definition 6.1

Local Maximum and local Minimum

A function f has a local maximum (or relative maximum) at c if there is an open interval (a,b) containing c such that $f(c) \geq f(x)$ for every $x \in (a,b)$.

Similarly, f has a local minimum at c if there is an open interval (a,b) containing c such that $f(c) \leq f(x)$ for every $x \in (a,b)$.

Definition 6.2

Absolute maximum and absolute minimum

A function f has an **absolute maximum** at c if $f(c) \geq f(x)$ for all x in domain of f. The number $f(c)$ is called maximum value of f in the domain. Similarly f has an **absolute minimum** at c if $f(c) \leq f(x)$ for all x in domain of f and the number $f(c)$ is called the minimum value of f on the domain. The maximum and minimum value of f are called extreme values of f.

NOTE

Absolute maximum and absolute minimum values of a function f on an interval (a,b) are also called the global maximum and global minimum of f in (a,b).

Criteria for local maxima and local minima

Let f be a differentiable function on an open interval (a,b) containing c and suppose that $f''(c)$ exists.

(i) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c.

(ii) If $f'(c) = 0$ and $f''(c) < 0$,then f has a local maximum at c.

NOTE

In Economics, if $y = f(x)$ represent cost function or revenue function, then the point at which $\frac{dy}{dx} = 0$, the cost or revenue is maximum or minimum.

Example 6.25

Find the extremum values of the function $f(x)=2x^3+3x^2-12x$.

Solution:

Given $f(x)= 2x^3+3x^2-12x$ … (1)

$f'(x) = 6x^2+6x-12$

$f''(x) = 12x + 6$

$f'(x)= 0 \Rightarrow 6x^2+6x-12 =0$

$\Rightarrow 6(x^2+x-2)= 0$

$\Rightarrow 6(x+2)(x-1)= 0$

$\Rightarrow x = -2 ; x = 1$

06_11th_BM-STAT_Ch-6-EM.indd 143 21-04-2020 12:22:00 PM